

The numodel package^{*}

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Abstract

A LuaLaTeX package for writing and rendering numerical models (Euler-integrated dynamical systems) directly inside LaTeX documents, aimed at physics teaching material. It provides a text-model pipeline (`\mvar`, `\mrule`, `\computemodel`), Forrester stock-and-flow diagrams, and optional plots of the computed time series via the sibling package `numodel-plot`.

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1 Introduction

`numodel` lets an author write a dynamical system (stocks, flows, helper variables, rules, and a stop condition) as a sequence of LaTeX macros and renders three complementary views of that model directly in the document:

- a *text model* — a typeset rule table with initial values (`\textmodel`);
- a *graphic model* — a Forrester stock-and-flow diagram with auto-layout (`\graphicmodel`);
- a *diagram* — a numerical Euler simulation plus PGFPlot of any pair of variables (`\computemodel` followed by `\diagrammodel`; the plot is rendered through the sibling package `numodel-plot`).

^{*}This document corresponds to `numodel` v0.9.0, dated September 6, 2026.

All three views are produced from a single set of declarations so the textbook description, the conceptual stock-and-flow diagram, and the numerical result of the same model are guaranteed to stay in sync. Variables and rules live in namespaces (*prefixes*) so a document can contain multiple independent models; `\newmodelprefix{P}` starts a fresh one. The simulation engine runs in Lua (through `luacode`) for $\mathcal{O}(1)$ appends and cheap min/max tracking; the rendering layer is pure `expl3`.

2 First example: a free-falling ball

A ball dropped from $h_0 = 100$ m under constant gravitational acceleration. The complete model:

```
\usepackage[syntax=EN]{numodel}

\newmodelprefix{ball}
\mvar{T}{t}{0}{\s}{2}{system}
\mvar{Dt}{dt}{0.1}{\s}{2}{system}
\mvar{V}{v}{0}{\m\per\s}{2}{stock}
\mvar{Y}{y}{100}{\m}{3}{stock}
\mvar{G}{g}{-9.81}{\m\per\s\squared}{3}{aux}

\mrule{V}{\ballV + \ballG * \ballDt}
\mrule{Y}{\ballY + \ballV * \ballDt}
\mrule{T}{\ballT + \ballDt}
\mstop{\ballY <= 0}
```

After the declarations above, three render commands produce the three views shown below, each from the *same* model.

2.1 `\textmodel` — rule table

The verbatim source rendered by `\textmodel`:

	model	initial values
1	$v = v + g \cdot dt$	$t = 0 \text{ s}$
2	$y = y + v \cdot dt$	$dt = 0.10 \text{ s}$
3	$t = t + dt$	$v = 0 \text{ m s}^{-1}$
4	IF $y \leq 0$ THEN STOP ENDIF	$y = 100 \text{ m}$ $g = -9.81 \text{ m s}^{-2}$

Each `\mvar` with a non-empty start value contributes a row in the *initial values* column; each `\mirule` contributes a row in the *model* column. Symbols come from the second `\mvar` argument (the display text), values are formatted through `siunitx`. `<=` is rendered as $\leq$.

2.2 `\graphicmodel` — Forrester diagram

`\graphicmodel` draws the same model as a stock-and-flow diagram. Stocks (type `stock`) are rectangles; auxiliaries (`aux`) and constants (`constant`) are circles, with two short horizontal dashes flanking the constant ring. In the default `tight` diagram-style, an `aux` or `constant` that is the direct inflow or outflow of a stock is absorbed into the valve label and not drawn as a separate node; `diagram-style=forrester` or `edu` keep both (Section 3).

Flows are inferred per additive term on the right-hand side of each stock rule. After dropping the self-carry term, every `+T` is a candidate inflow and every `-T` a candidate outflow of the target stock; the first non-system, non-`Dt` variable appearing inside `T` becomes the flow variable (`aux` and `stock` beat `constant` when several candidates coexist). When the same flow variable surfaces as an outflow term of stock A and a matching inflow term of stock B, the pair is promoted to a *between-flow*:

one valve drains A into B instead of two separate valves. If the flow variable of an inflow term is itself a stock, the source stock takes the valve's place – with a matching outflow term on its own rule (a conserved-quantity between-flow) or, when no such match exists, with a cloud-fed phantom valve plus a causal link to the source stock:



Layout is automatic from the rule graph: stocks (with their inflow/outflow valves) sit on the bottom row, auxiliaries on the middle row, and constants on the top row, each filled left-to-right in declaration order. The order of `\mvar` calls therefore drives the horizontal reading order of the diagram – reordering, inserting or holding back a `\mvar` nudges the visible arrangement without touching any grid coordinates, which is the primary lever a document author has over the look of the diagram. Manual placement remains available through `gridx/gridy` keys on the `\mvar` call (see Section 4). Wide diagrams can be capped to a maximum number of grid columns with `\numodelsetup{grid_maxx=N}`: when a row reaches N, the auto layout shifts the affected items up one row and continues filling.

2.3 `\computemodel + \diagrammodel` — numerical plot

`\computemodel` iterates the rules forward in time using Euler integration with step size `\ballDt`, stopping when `\mstop`'s condition becomes true (or when the `maxiter` safety limit is reached, see Section 3). `\diagrammodel{xvar}{yvar}{label}` then plots one variable against another:

```
\computemodel
\diagrammodel{T}{Y}{ball-fall}
```

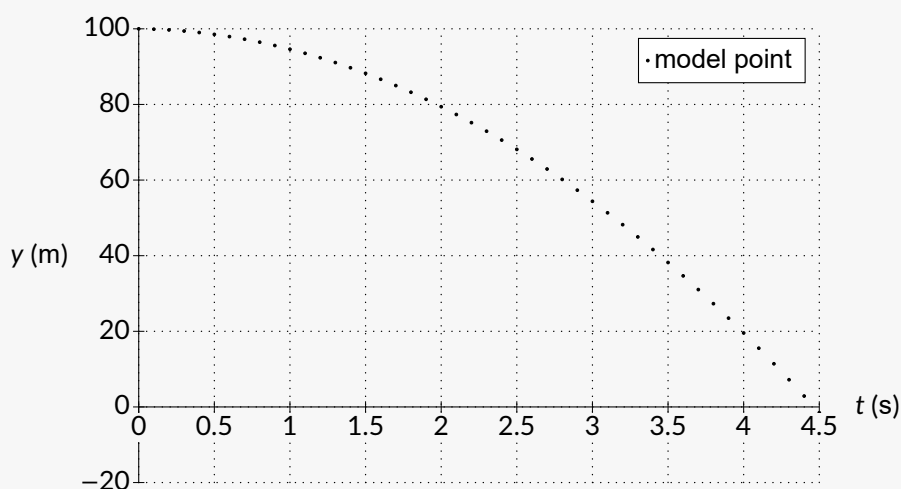


Figure 1: $y(t)$ -diagram

Axis ranges, tick lattice, and labels are computed automatically from the simulated min/max of each variable; the plot inherits the `numodel-plot` style (see that package's documentation for configuration).

3 Configuration

`\numodelsetup` Runtime configuration:

```
\numodelsetup{syntax=NL, maxiter=50000}
```

The same keys can also be passed as package options: `\usepackage[syntax=NL]{numodel}`. Recognised keys:

syntax Language tag for the rule-table rendering. Built-in values:

EN (default) XMI-style ALL-CAPS keywords: IF/THEN/ELSE, AND, OR, ABS, SIGN, ...

NL Dutch CoachTaal keywords: Als/Dan/Anders, EN, OF, Abs, Teken, ...

FPEVAL (aliases `fpeval`, `l3fp`) shows each rule body verbatim in `\l3fp` (`\fpeval`) syntax, exactly as typed in `\mrule` – lowercase function names, literal `*`, `^`, `&`, `?`: – typeset monospace. Variable macros still resolve to their display names; the ternary is *not* expanded into IF/THEN/ELSE and the stop row reads `stop: <cond>`. Decimal mark defaults to point.

Each language tag `X` corresponds to a file `numodel-X.def` located via `kpse` when the package processes the key. The package ships with `numodel-EN.def` and `numodel-NL.def`; drop your own `numodel-FR.def` (or any other tag) in `TEXMFHOME/tex/latex/numodel/` and select it with `\usepackage[syntax=FR]{numodel}` – no package rebuild needed. The setting affects display only; the expression syntax in `\mrule` bodies is always `\fp_eval-compatible`.

maxiter Safety limit on the number of `\computemodel` iterations (default 20 000). When reached, the simulation aborts with a warning naming the unmet stop condition.

graphscalex Horizontal grid spacing in centimetres for `\graphicmodel`'s Forrester layout (default 2). Larger values spread the diagram out horizontally.

graphscaley Vertical grid spacing in centimetres for `\graphicmodel`'s Forrester layout (default 2). Larger values spread the diagram out vertically.

`graphscalex/graphscaley` control only the empty space between diagram elements: the nodes themselves (stocks, valves, auxiliaries, constants) have fixed dimensions expressed in `em` and are not individually configurable. Because `em` is relative to the current font size, the way to grow or shrink the nodes uniformly is to change the font size around the `\graphicmodel` call – e.g. wrap it in `{\small \graphicmodel}` or `{\large \graphicmodel}`. Pick spacing with `graphscalex/graphscaley`, pick node size with the surrounding font.

gridmaxx Maximum number of grid columns the auto layout may fill on any one row before wrapping (integer, default 0 = no limit). When the limit is reached, items already placed on the affected row (and everything above it for the stocks row, everything but stocks for the aux row, only the constants for the constants row) shift up by one row to free space, and placement continues from column 0. Manually positioned variables (`\mvar[gridx=...,gridy=...]`) are kept where they are. When wrapping is active the default centring of the aux row and the right-aligning of the stocks row are disabled, so the diagram fills left-to-right, bottom-to-top.

diagram-style Rendering style for the case where a helper or constant is the direct inflow/outflow of a stock. Three values:

tight (default) the valve takes the helper's/constant's label; the helper/constant itself is not drawn as a separate node. Compact and most LaTeX-native.

forrester Forrester/Sterman convention: the valve is drawn without a label and the helper/constant remains as a separate node connected to the valve by a causal arrow.

edu Didactic dual form: the valve carries the label *and* the helper/constant is drawn as a separate node with a causal arrow to the valve. Visually busy but pedagogically explicit.

flowarrow-style Visual style of the flow pipe. `hollow` renders the classic Forrester double-line pipe with an open arrow head; `filled` renders a thick solid arrow. The default tracks `diagram-style`: `forrester` picks `hollow`, the other styles pick `filled`. An explicit value overrides this coupling.

valve-style Visual style of the valve node. `valve` draws the bow-tie/butterfly icon (Forrester); `circle` draws an empty circle on the flow pipe; `edu` draws a labelled circle (the flow variable's display text inside). The default tracks `diagram-style`: `forrester` picks `valve`, the other styles pick `edu`.

flowarrow-cloud-tip Whether the open end of an inflow or outflow pipe is anchored to a cloud node, signalling the model boundary. Default tracks `diagram-style`: `forrester` picks `true`, the other styles pick `false`. May be set globally via `\numodelsetup`, per-render via `\graphicmodel`, or per-stock via `\mvar[flowarrow-cloud-tip=...]`. The most specific source wins.

units Whether the *initial values* cells in `\textmodel` display the SI unit alongside the value (`\qty`) or only the numeric value (`\num`). Boolean, default `true`. May also be supplied to `\textmodel` as `el[units=false]` as a per-table override; the global setting is restored after rendering.

tblrenv Which tabularray environment wraps the `\textmodel` table. Three values: `longtblr` (default, page-breakable); `tblr` (inline, no page breaks); `talltblr` (inline, no page breaks, supports `\caption/notes`). Pick `tblr` or `talltblr` when `\textmodel` sits inside an enclosing environment that suppresses page breaks (`subfigure`, `minipage`, ...); the default `longtblr` would otherwise emit spurious “(Continued)” / *Continued on next page* markers in that setting. May also be supplied per render as `\textmodel[tblrenv=...]`; the global setting is restored afterwards.

decimal-separator Decimal mark used by every number that `numodel` renders: the *initial values* column of `\textmodel`, and the tick labels of `\diagrammodel`. Two values:

comma use a comma (`siunitx output-decimal-marker={,}`, `pgfplots/pgf/number for mat/use comma`).

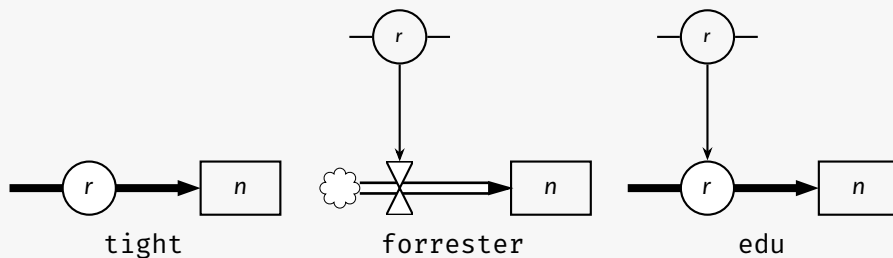
point use a full stop (`siunitx output-decimal-marker={.}`, `pgfplots/pgf/number for mat/use period`).

The default tracks syntax: NL picks comma, EN and FPEVAL pick point. Other language files can publish a default by defining `__numodel_kw_LANG_dsep_default`: (expanding to point or comma); when the macro is absent the default is point. An explicit `decimal-separator` key locks that choice and overrides any future syntax change. The override is scoped: `numodel` applies it only inside its own renderers (`siunitx`’s state is restored on group exit), so a document-wide `\sisetup` is not perturbed.

3.1 Diagram styles in practice

The same model rendered under each of the three diagram-style values. The model is the simplest case that distinguishes the styles: one stock *N* with a constant inflow *R*:

```
\newmodelprefix{flux}
\mvar{T}{t}{0}{s}{2}{system}
\mvar{Dt}{dt}{1}{s}{2}{system}
\mvar{N}{n}{0}{s}{0}{stock}
\mvar{R}{r}{5}{per}{s}{2}{constant}
\mrule{N}{\fluxN + \fluxR * \fluxDt}
\mrule{T}{\fluxT + \fluxDt}
\mstop{\fluxT >= 5}
\begin{tabular}{@{}ccc@{}}
\graphicmodel[diagram-style=tight] &
\graphicmodel[diagram-style=forrester] &
\graphicmodel[diagram-style=edu] \\
\texttt{tight} & \texttt{forrester} & \texttt{edu}
\end{tabular}
```



- `tight` collapses the constant *R* into the valve label, producing the most compact diagram.
- `forrester` keeps the canonical System-Dynamics convention: unlabelled bow-tie valve, the constant remains a separate node, the link from *R* to the valve is a thin causal arrow.

- edu is a didactic dual: the valve carries the label *and* the constant remains as a separate node with a causal arrow. Less compact but pedagogically explicit – useful when first introducing the stock/flow vocabulary.

4 Public API

4.1 Variables and rules

`\mvar` Declares a model variable. Signature:

```
\mvar[<keys>]{<Name>}{<text>}{<start>}{<unit>}{<sig>}{<type>}
```

where *<Name>* is a short alphabetic identifier (the prefix-qualified accessor becomes `\<prefix><Name>`, along with a family of suffixed accessor macros `\<prefix><Name><suffix>` documented in Section 6), *<text>* is the math-mode display symbol used in the rule table and diagram (e.g. F_{res}), *<start>* is the initial value (a number, or empty for helpers computed by a rule, or any `expl3` fp-evaluable expression involving previously defined model variables), *<unit>* is a bare `siunitx` unit macro sequence (e.g. `\m\per\s\squared`), *<sig>* is the number of significant figures used by `\<prefix><Name>`, `num/qty`, and *<type>* is one of `stock`, `aux`, `constant`, or `system`. Each English type also accepts a Dutch alias (`voorraad`, `hulp`, `constante`, `systeem`) for backwards compatibility with existing teaching material. See Section 5.

Naming caveat. All accessor macros of a given prefix share one flat TeX namespace: `\mvar{X}` generates `\<prefix>X` and every `\<prefix>X<suffix>` listed in Section 6. Pick *<Name>*s so that no name equals another name followed by such a suffix, otherwise the two definitions silently overwrite each other. The canonical trap is declaring both `T` and `Tmax`: the `max`-suffix accessor of `T` (the extremum `\<prefix>Tmax`) collides with the no-suffix accessor of `Tmax`, so `\<prefix>Tmax` inside a rule expression ends up meaning whichever was declared last. The same hazard applies to prefix choices – avoid prefixes where one is a concatenation of another with a variable name (e.g. `ball` with a variable `Y` versus a prefix `ballY`). An `\mvar` name of `steps` would similarly collide with the `per`-prefix iteration accessor `\<prefix>steps` set by `\computemodel`.

Optional *<keys>*:

prefix Override the current prefix for this single call.

gridx, gridy Manual placement in the `\graphicmodel` grid; integers, `-1` leaves the slot to auto-layout (default).

alias Math-mode token list that replaces the entire *initial values* cell.

aliasleft, aliasright Replace just the left symbol or right value half of the cell.

`\mrule` Adds a rule of the form $\langle LHS \rangle \leftarrow \langle expr \rangle$. Signature:

```
\mrule*[<keys>]{<LHS>}{<expr>}
```

Both forms add the rule to *both* the rule table (`\textmodel`) and the simulation (`\computemodel`); execution is identical. The star only changes the typeset layout when *<expr>* is a ternary cond ? a : b. Without the star the ternary is rendered on a single table row,

```
IF cond THEN lhs = a ELSE lhs = b ENDIF
```

which is compact but wide. With the star (`\mrule*`) the same ternary is broken across rows,

```
IF cond THEN
  lhs = a
ELSE
  lhs = b
ENDIF
```

keeping the table column narrow so a `\graphicmodel` can sit alongside it, and making the source itself easier to read. For non-ternary expressions the star has no effect. *<expr>* may use the full `\fp_eval` expression grammar. See Table 2 (Section 4.2) for a complete overview of supported operators and functions with their XMILE and CoachTaal equivalents.

`\mruletext` Inserts a free-text row in the rule table without registering a rule with the simulator. Signature:

`\mruletext[<keys>]{<text>}`. Useful for inserting comments or section dividers in long rule tables.

`\mstop` Sets the simulation stop condition. Signature: `\mstop[<keys>]{<expr>}`. The simulation halts at the first step where `<expr>` evaluates true. Exactly one `\mstop` per model prefix is required before `\computemodel`. Without one, `\computemodel` issues a warning.

4.2 Expression reference

Table 2 lists all expression constructs for `\mrule` bodies. The **XMILE** column shows XMILE v1.0 syntax, **I3fp** shows the `\fp_eval`-compatible form used inside `\mrule`, and **CoachTaal** shows the Coach 7 equivalent (function names from the standard CoachTaal math reference; note that argument separators in CoachTaal are semicolons). **Orange** entries are not yet fully supported by numodel (the expression computes correctly, but the rendered keyword in the rule table is not yet translated). *Italic* cells contain a derived equivalent rather than a native keyword.

Table 2: Expression reference: XMILE, `\fp_eval`, and CoachTaal

XMILE	I3fp (<code>\fp_eval</code>)	CoachTaal
<i>Arithmetic operators</i>		
+	+	+
-	-	-
*	*	*
/	/	/
^	^	^
MOD(x, y)	<i>x - trunc(x/y)*y</i>	<i>x - Entier(x/y)*y</i>
<i>Comparison operators</i>		
<	<	<
<=	<=	<=
>	>	>
>=	>=	>=
=	=	=
<>	!=	<>
<i>Boolean operators</i>		
AND	&&	EN
OR		OF
NOT	!	NIET
<i>Control flow</i>		
IF c THEN a ELSE b	c ? a : b	Als c Dan a Anders b EindAls
<i>General math functions</i>		
ABS(x)	abs(x)	Abs(x)
SIGN(x)	sign(x)	Teken(x)
SQRT(x)	sqrt(x)	Sqrt(x)
INT(x)	trunc(x)	<i>Entier(Abs(x))*Teken(x)</i>
INT(x+0.5)	round(x)	<i>Round(x)</i>
INT(x)	floor(x)	Entier(x)
-INT(-x)	ceil(x)	<i>-Entier(-x)</i>
MIN(x, y)	min(x, y)	Min(x; y)
MAX(x, y)	max(x, y)	Max(x; y)
—	fact(x)	Fac(x)
INT(LOG10(ABS(x)))	logb(x)	<i>Entier(Log(Abs(x)))</i>
<i>Exponential and logarithmic</i>		
EXP(x)	exp(x)	Exp(x)
LN(x)	ln(x)	Ln(x)

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Table 2: Expression reference: XMILE, \fp_eval, and CoachTaal (Continued)

XMILE	l3fp (\fp_eval)	CoachTaal
LOG10(x)	<i>ln(x)/ln(10)</i>	Log(x)
<i>Trigonometry (radians)</i>		
SIN(x)	sin(x)	Sin(x)
COS(x)	cos(x)	Cos(x)
TAN(x)	tan(x)	Tan(x)
ARCSIN(x)	asin(x)	Arcsin(x)
ARCCOS(x)	acos(x)	Arccos(x)
ARCTAN(x)	atan(x)	Arctan(x)
ARCTAN2(y,x)	atan(y,x)	Arctan(y/x)
1/TAN(x)	cot(x)	1/Tan(x)
1/SIN(x)	csc(x)	1/Sin(x)
1/COS(x)	sec(x)	1/Cos(x)
ARCSIN(1/x)	acsc(x)	Arcsin(1/x)
ARCCOS(1/x)	asec(x)	Arccos(1/x)
ARCTAN(1/x)	acot(x)	Arctan(1/x)
ARCTAN2(x,y)	acot(y,x)	Arctan(x/y)
<i>Trigonometry (degrees)</i>		
SIN(PI/180*x)	sind(x)	Sin(Pi/180*x)
COS(PI/180*x)	cosd(x)	Cos(Pi/180*x)
TAN(PI/180*x)	tand(x)	Tan(Pi/180*x)
180/PI*ARCSIN(x)	asind(x)	180/Pi*Arcsin(x)
180/PI*ARCCOS(x)	acosd(x)	180/Pi*Arccos(x)
180/PI*ARCTAN(x)	atand(x)	180/Pi*Arctan(x)
180/PI*ARCTAN2(y,x)	atand(y,x)	180/Pi*Arctan(y/x)
1/TAN(PI/180*x)	cotd(x)	1/Tan(Pi/180*x)
1/SIN(PI/180*x)	cscd(x)	1/Sin(Pi/180*x)
1/COS(PI/180*x)	secd(x)	1/Cos(Pi/180*x)
180/PI*ARCSIN(1/x)	acscd(x)	180/Pi*Arcsin(1/x)
180/PI*ARCCOS(1/x)	asecd(x)	180/Pi*Arccos(1/x)
180/PI*ARCTAN(1/x)	acotd(x)	180/Pi*Arctan(1/x)
180/PI*ARCTAN2(x,y)	acotd(y,x)	180/Pi*Arctan(x/y)
<i>Constants</i>		
PI	pi	Pi
e	exp(1)	Exp(1)
INF	inf	—
NAN	nan	—
PI/180	deg	Pi/180
1	true	Aan
0	false	Uit
<i>Simulation-specific functions (XMILE only)</i>		
TIME	<i>user variable (\mvar)</i>	<i>user variable</i>
DT	<i>user variable (\mvar)</i>	<i>user variable</i>
STEP(h, t ₀)	<i>T >= t₀ ? h : 0</i>	<i>Als T >= t₀ Dan h Anders 0 EindAls</i>
RAMP(s, t ₀)	<i>T > t₀ ? s*(T-t₀) : 0</i>	<i>Als T > t₀ Dan s*(T-t₀) Anders 0 EindAls</i>
DELAY(x, dt)	<i>not supported</i>	<i>not supported</i>
SMOOTH(x, t)	<i>not supported</i>	<i>not supported</i>
INIT(x)	<i>not supported</i>	<i>not supported</i>
PREVIOUS(x)	<i>not supported</i>	<i>not supported</i>

Continued on next page

Table 2: Expression reference: XMILE, \fp_eval, and CoachTaal (Continued)

XMILE	l3fp (\fpeval)	CoachTaal
<i>RANDOM(0,1)</i>	<i>rand()</i>	<i>Rand</i>
<i>RANDOM(lo,hi)</i>	<i>lo + (hi-lo)*rand()</i>	<i>lo + (hi-lo)*Rand</i>
<i>not supported</i>	<i>randint(n)</i>	<i>not supported</i>
<i>not supported</i>	<i>randint(m,n)</i>	<i>not supported</i>

4.3 Render commands

- `\textmodel` Renders the rule-and-startvalue table. Optional [`<keys>`] accepts `prefix=<name>` (render a non-current model), `units=true` or `units=false`, and `tblrenv=tblrlongtblrtalltblr`. Each per-call key overrides the global `\numodelsetup` setting for this single render only; the global state is restored afterwards.
- `tblrenv` selects which `tabularray` environment wraps the table: `longtblr` (default) breaks across pages, `tblr` renders inline without page breaks, `talltblr` renders inline but with caption and note support. Use `tblr` or `talltblr` when `\textmodel` sits inside an enclosing environment that suppresses page breaks (`subfigure`, `minipage`, ...); the default `longtblr` would otherwise emit spurious continuation markers there.
- Row spacing.** `numodel` loads `tabularray` and sets `\SetTblrInner{rowsep=0pt}` globally so the rule listing renders compactly. This applies to *every* `tabularray` table in the document; if you want the default spacing back in your own tables, issue `\SetTblrInner{rowsep=2pt}` (or whatever value you prefer) somewhere in your document.
- `\graphicmodel` Renders the Forrester stock-and-flow diagram. Variables of type `stock` become rectangles, `c`, `onstant` become circles, `aux` become identifier nodes; flow arrows connect stocks to constant or helper sources/sinks based on which variables appear in the right-hand side of stock-updating rules.
- Optional [`<keys>`] accepts `prefix=<name>` (render a non-current model) and `diagramstyle=tightforresteredu`, which overrides the global `\numodelsetup` setting for this single render only. The global state is restored afterwards, so multiple `\graphicmodel` calls can each pick their own style without re-issuing `\numodelsetup`.
- `\computemodel` Runs the Euler simulation. Each iteration step is executed in LaTeX itself: `expl3`'s `\fp_eval` (the engine behind the LaTeX2e `\fpeval`) evaluates the stop condition and then every `\mrule` body in declaration order, writing each new value into the accessor macro `\<prefix><Name>`. The per-step values are appended to Lua tables on the side – storing the time series as growing TeX token lists would cost $O(N)$ per append and $O(N^2)$ overall, whereas the Lua-table append is $O(1)$, so a 20 000-step run stays linear in total work. Lua also keeps the running min and max so no second pass over the series is needed. After `\computemodel` returns, `\<prefix><Name>min` / `\<prefix><Name>max` hold the extrema, `\<prefix><Name>` holds the final-step value, `\<prefix>steps` expands to the number of recorded samples (i.e. the iteration count N), and the full time-series can be retrieved with `\mcoords` / `\mstep`.
- `\diagrammodel` Convenience wrapper that produces a complete figure with caption and label. Signature:
- `\diagrammodel[<keys>]{<xvar>}{<yvar>{,...}}[<extra>]{<label>}`
- Reads min/max and display text from `\<prefix><xvar>` etc., delegates to `\drawplot` from `numodel-plot`, and emits `\caption{$<prefix><yvar><text>(<prefix><xvar><text>)$-diagram}\label{fig:<label>}`. The optional `<extra>` argument is appended to the axis body, useful for additional `\addplot` lines (annotations, theoretical curves). Must be called after `\computemodel`.
- The `<yvar>` argument accepts a comma-separated list of variables: every entry whose `unitraw` matches the first one's is drawn into the same diagram as discrete model points (no connecting lines), with the y-axis range scaled to the joint min/max of the kept series. Entries with a non-matching unit are dropped and a warning is issued. The seven colours cycle through Okabe & Ito's colour-blind safe palette (yellow omitted), ordered so consecutive series differ in luminance – which keeps them distinguishable on a greyscale printout as well. The legend lists each kept variable's display text.
- The `prefix=` key likewise accepts a comma-separated list, so the same y-variables are plotted across several models in one diagram – useful for direct visual comparison between alternative model variants of the same physical setup. Series order is prefix-major (all y-variables of the first prefix come before any of the second), the colour cycle continues across the cross product, and axis ranges reduce over the union of all (prefix, yvar) pairs. The unit filter is applied against the first prefix's first y-variable, so the same short names are assumed to share unit and display text across prefixes. When more

than one prefix is supplied the prefix is appended in parentheses after each y-variable's display text in the legend, e.g. U_3 (ptc) and U_3 (pw). Single-prefix calls render exactly as before.

`\diagrammodel[prefix={ptc,pw}]{T}{U_3,U_par}{compare-models}`

Example – two quantities in one diagram. The free-fall ball from the introduction is extended with gravitational potential energy $E_p = m g h$ (written as $-m G Y$ because $G = -9,81$ already encodes the downward direction) and kinetic energy $E_k = \frac{1}{2} m v^2$. Both share the unit `\joule`, so the multi-series filter keeps both and scales the y-axis to their combined range. At $t = 0$ all energy sits in E_p ; as the ball falls, E_p drops and E_k grows. The two aux variables receive a start-value expression so that step 0 is recorded with the physically meaningful initial energies rather than empty values.

```
\newmodelprefix{nrg}
\mvar{T}{t}{0}{\s}{2}{system}
\mvar{Dt}{dt}{0.1}{\s}{2}{system}
\mvar{V}{v}{0}{\m\per\s}{2}{stock}
\mvar{Y}{y}{100}{\m}{3}{stock}
\mvar{m}{m}{0.5}{\kg}{3}{constant}
\mvar{G}{g}{-9.81}{\m\per\s\squared}{3}{constant}
\mvar{Ep}{E_p}{-\nrgm * \nrgG * \nrgY}{\joule}{3}{aux}
\mvar{Ek}{E_k}{0.5 * \nrgm * \nrgV * \nrgV}{\joule}{3}{aux}
\mrule{V}{\nrgV + \nrgG * \nrgDt}
\mrule{Y}{\nrgY + \nrgV * \nrgDt}
\mrule{Ep}{-\nrgm * \nrgG * \nrgY}
\mrule{Ek}{0.5 * \nrgm * \nrgV * \nrgV}
\mrule{T}{\nrgT + \nrgDt}
\mstop{\nrgY <= 0}
\computemodel
\diagrammodel{T}{Ep,Ek}{ball-energy}
```

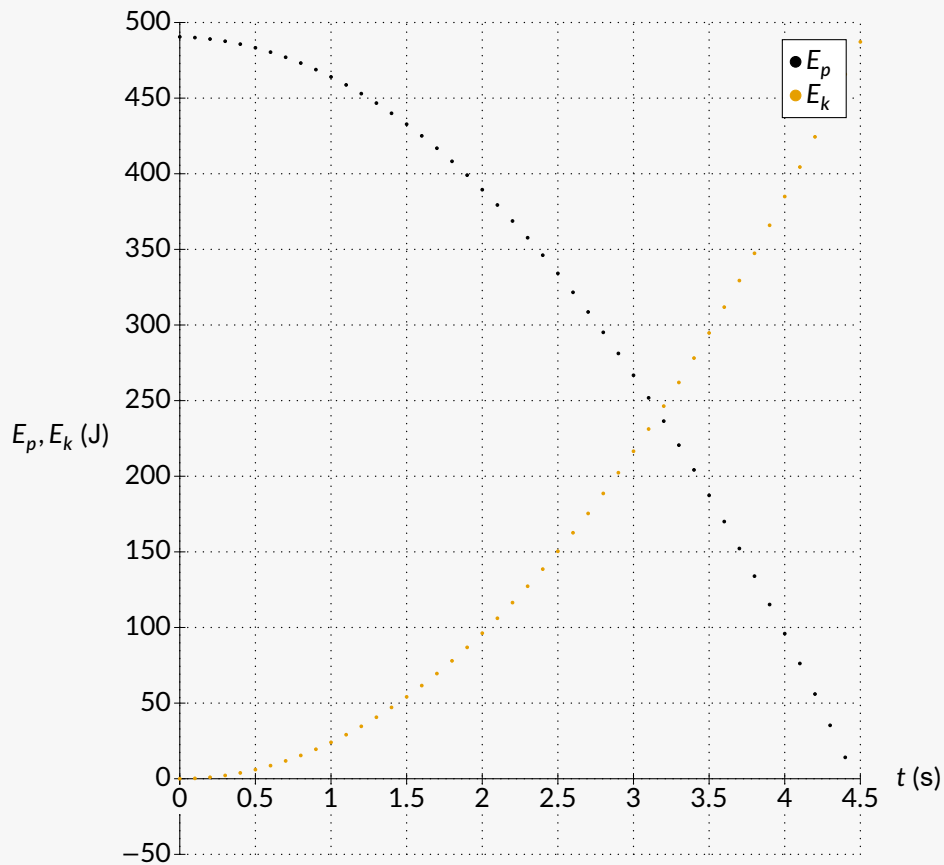


Figure 2: $E_p, E_k(t)$ -diagram

4.4 Series accessors

`\mcoords` Returns a comma-separated PGFPlots coordinate list of the simulated series for two variables. Fully expandable. Signature: `\mcoords{<xvar>}{<yvar>}` (current prefix); `\mcoordsp{<prefix>}{<xvar>}{<yvar>}` (explicit prefix). Both forms are usable inside `\addplot` coordinates{... }.

`\mstep` Returns the value of one variable at a chosen iteration. Fully expandable. Signature: `\mstep{p{<Name>}{<i>}}`; `\mstepp{<prefix>}{<Name>}{<i>}`. The current prefix is prepended automatically by `\mstep`. Step indexing is 0-based: step 0 is the initial-values row, step $N-1$ is the final recorded step after `\computemodel`. Negative indices count from the end (Python-style), so `\mstep{Y}{-1}` is the last recorded y and `\mstep{Y}{-2}` is the penultimate one. Raises a `nu\` model error when $\langle i \rangle$ falls outside the recorded range $(0..\langle \text{prefix} \rangle \text{steps} - 1)$ for non-negative indices, or $-\langle \text{prefix} \rangle \text{steps}..-1$ for the Python-style negative form), which surfaces typo'd indices and "ran N steps but indexed step N " off-by-one mistakes rather than expanding to nothing and propagating empty arguments into the surrounding plot.

Example – a red secant line through the last two simulated (t, y) points of the free-fall ball (the model from the first example), extrapolated across the whole t -domain and labelled in the legend. The slope is the rise-over-run of the last two steps, the intercept is the final y -value. Inside the optional [`<extra>`] argument `\mstep` expands as a literal number into PGFPlots' math parser, so each `\mstep` is evaluated only once (when the expression is constructed) rather than per sample. Applied to the `ball` model:

```
\diagrammodel{T}{Y}{%
  \addplot[red, very thick, domain=\ballTmin:\ballTmax]
    {\mstep{Y}{-1}
     + (\mstep{Y}{-1} - \mstep{Y}{-2})
     / (\mstep{T}{-1} - \mstep{T}{-2})
     * (x - \mstep{T}{-1})};
  \addlegendentry{secant endpoints}
]{ball-fall-secant}
```

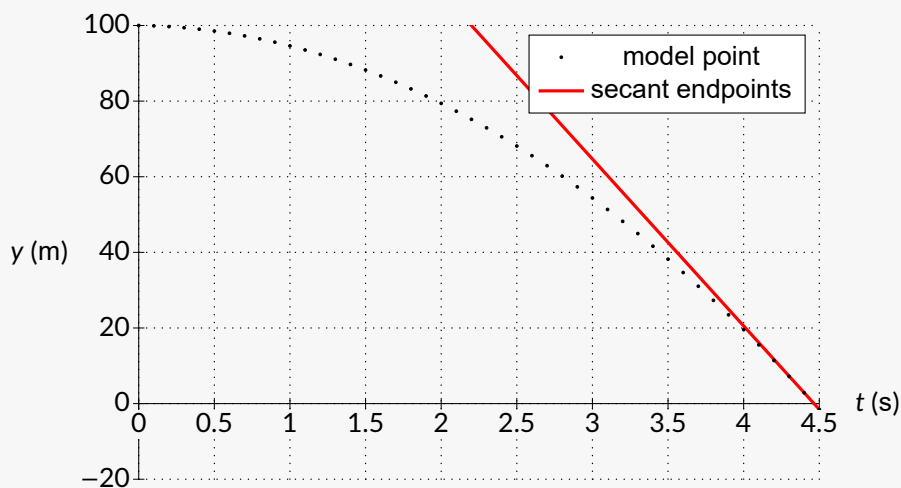


Figure 3: $y(t)$ -diagram

4.5 Namespace management

`\newmodelprefix` Creates a new model namespace and switches to it. Signature: `\newmodelprefix{<name>}`. Subsequent `\mvar` / `\mrule` / `\mstop` calls bind to this prefix. All generated accessors are prefixed (so `\mvar{Y}{y}{...}{...}{...}{...}` under prefix `ball` produces `\ballY`, `\ballYtext`, etc.).

`\switchmodelprefix` Switches to a previously created prefix. Useful when a document defines several models early and renders them later out of order. Signature: `\switchmodelprefix{<name>}`.

`\NumodelForEachVar` Iterates a token list over every registered variable across every known prefix. Signature: `\NumodelForEachVar`

`odelForEachVar{<code with #1>};` inside the body, #1 is the full prefixed name (e.g. `ballY`). Used by external tools (e.g. a worksheet system) that need to expand every model accessor.

4.6 Shared valves

When the same auxiliary variable appears as the inflow of more than one stock, `\graphicmodel` draws a single valve next to the first such stock and threads a curved branch (`to[bend left=3, 0]` – the same bend as the curved causal arrows) from that valve over the primary stock into each additional one. The auto-layout leaves the involved stocks side by side on the same row so the branch stays compact. A minimal example:

```
\mvar{R}{r}{5}{\m\cubed\per\s}{2}{aux}
\mvar{Va}{V_1}{0}{\m\cubed}{3}{stock}
\mvar{Vb}{V_2}{0}{\m\cubed}{3}{stock}
\mrule{Va}{\Va + \R * \Dt}
\mrule{Vb}{\Vb + \R * \Dt}
```

renders one R valve, a straight pipe $R \rightarrow V_a$, and a curved branch $R \rightarrow V_b$ arcing over V_a . (Note: TeX control words do not accept digits, so the macro names V_a/V_b are used internally while the display texts V_1/V_2 appear in the rendered diagram and table.)

Shared *outflows* are handled symmetrically. When one variable drains more than one stock, the valve is placed to the right of the last-declared source stock, and every earlier source attaches with a curved branch arcing over the intervening stocks into the shared valve. This matters in particular for physical models where a single Stefan–Boltzmann constant or friction coefficient drains several reservoirs into the same sink.

Two flow-classification refinements support these layouts:

- The flow-variable heuristic prefers `aux` and `stock` variables over constants when both appear in the same additive term, so an inflow valve carries the meaningful rate variable instead of an earlier-declared scaling constant.
- Top-level parenthesised inflow terms of the form $(A - B) * C$ are distributed into $+A * C$, $-B * C$ before classification, so A surfaces as the inflow valve and B (typically a radiative or dissipative loss involving a constant) surfaces as the outflow valve.

5 Variable types

The sixth `\mvar` argument (*<type>*) tags a variable with a role. The type drives both `\textmodel` layout (whether the row appears in *initial values*) and `\graphicmodel` node shape. Each type has a canonical English name and a Dutch alias; the two are interchangeable.

stock A stock that accumulates over time. Drawn as a rectangle in the Forrester diagram; its rule must be of the form $\text{stock} \leftarrow \text{stock} + \dots$ so that the integrator can detect inflows and outflows. Receives an *initial values* row.

constant A constant parameter (mass, gravitational acceleration, spring stiffness, ...). Drawn as a circle. Receives an *initial values* row.

aux An auxiliary variable computed from other variables on each step. Drawn as a plain identifier node, no rectangle. No *initial values* row (start value is normally left empty).

system System-level bookkeeping (time T , step size Dt , terminal time, ...). Drawn separately, no flow arrows. Receives an *initial values* row.

6 Generated accessors

Each `\mvar` call generates a family of accessor macros named `\<prefix><Name><suffix>`. The full set:

(*no suffix*) Current numeric value (post-rule-update if inside a step, post-final-step after `\computem_odel`).

text The display symbol passed as the second argument of `\mvar`.

unit `\unit{...}` applied to the unit argument (siunitx-formatted).

unitraw The unit argument verbatim, without the `\unit{}` wrapper, suitable for use as a building block (e.g. `\xlabelunit` in `numodel-plot`).

num The current value formatted via siunitx's `\num{}` with the variable's significant figures. Used by `\textmodel` in the *initial values* column when `units=false`.

qty As `num`, but including the unit (`\qty{value}{unit}`). Used by `\textmodel` in the *initial values* column when `units=true` (the default).

pre As `qty`, but with engineering prefix mode (e.g. `1500 W` renders as $1{,}5\text{ k}\text{W}$).

sign Significant-figure count (raw integer).

type Variable type (raw string).

min, max Extrema over the simulated series. Empty before `\computemodel`.

gridx, gridy Manual placement coordinates in the Forrester grid; -1 if left to auto-layout.

alias, aliasleft, aliasright Override tokens for the *initial values* cell layout. `aliasright` replaces just the value part of the cell (keeping the symbol and equals sign); `aliasleft` replaces just the symbol; `alias` replaces the entire cell.

In addition, `\computemodel` writes one per-prefix accessor (no variable name in the middle):

<prefix>steps The number of recorded samples after the most recent `\computemodel` call for this prefix, i.e. the iteration count N . Empty (undefined) before the first `\computemodel`. Picking steps as an `\mvar` name would collide with this accessor – see the naming caveat in Section 4.

Example – the alias keys exist primarily as a worksheet hook: the same model can be rendered as a fully-specified reference *and* as a fill-in-the-blanks exercise. `aliasright=\cdots` blanks the value while keeping the symbol visible (asking *what value* belongs there), `alias={?}` blanks the entire cell (asking the student to supply both the symbol and the value from physical knowledge). Crucially, `\computemodel` keeps using the original numeric start value, so the diagram or computed answer for the rest of the model remains correct – only the rendered table changes. The row for y now reads $y = \dots$ (asking the student to identify the initial height from a graph or photograph) and the row for g shows just $?$ (asking the student to supply the gravitational acceleration from memory). The rule column is unaffected – `\mrule` has its own `alias/aliasright` keys for blanking rule bodies if needed.

```
\newmodelprefix{quiz}
\mvar{T}{t}{0}{\s}{2}{system}
\mvar{Dt}{dt}{0.5}{\s}{2}{system}
\mvar{V}{v}{0}{\m\per\s}{2}{stock}
\mvar[aliasright=\cdots]{Y}{y}{50}{\m}{3}{stock}
\mvar[alias={?}]{G}{g}{-9.81}{\m\per\s\squared}{3}{constant}
\mrule{V}{\quizV + \quizG * \quizDt}
\mrule{Y}{\quizY + \quizV * \quizDt}
\mrule{T}{\quizT + \quizDt}
\mstop{\quizY <= 0}
\textmodel
```

	model	initial values
1	$v = v + g \cdot dt$	$t = 0 \text{ s}$
2	$y = y + v \cdot dt$	$dt = 0.50 \text{ s}$
3	$t = t + dt$	$v = 0 \text{ m s}^{-1}$
4	IF $y \leq 0$ THEN STOP ENDIF	$y = \dots$?

7 Multiple models in one document

Each `\newmodelprefix` starts a fresh namespace. This lets a document contain several unrelated models without name clashes:

```

\newmodelprefix{ball}
\mvar{Y}{y}{100}{\m}{3}{stock}
% ... ball model ...

\newmodelprefix{spring}
\mvar{X}{x}{0.1}{\m}{3}{stock}
% ... spring model ...

% Render the ball model:
\switchmodelprefix{ball}\textmodel\computemodel\diagrammodel{T}{Y}{fall}

% Render the spring model:
\switchmodelprefix{spring}\textmodel\computemodel\diagrammodel{T}{X}{spring}

```

Each prefix carries an independent rule list, stop condition, and recorded series.

8 Requirements

numodel requires LuaLaTeX (the engine, for the Lua runtime) and TeX Live 2022 or later. Mandatory dependencies: `expl3`, `xparse`, `l3keys2e`, `amsmath`, `amssymb`, `xfrac`, `tikz`, `luacode`, `siunitx`, `float`, and the sibling package `numodel-plot` (which itself pulls in `pgfplots`). The companion Lua module `numodel.lua` must be installed alongside the `.sty` in a directory searched by `kpse`.